Kernix 0.1.1 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

05 workloads

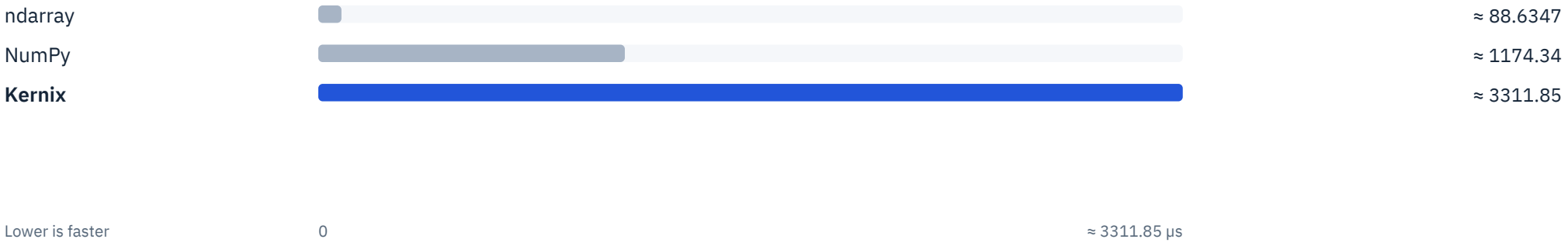
15 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

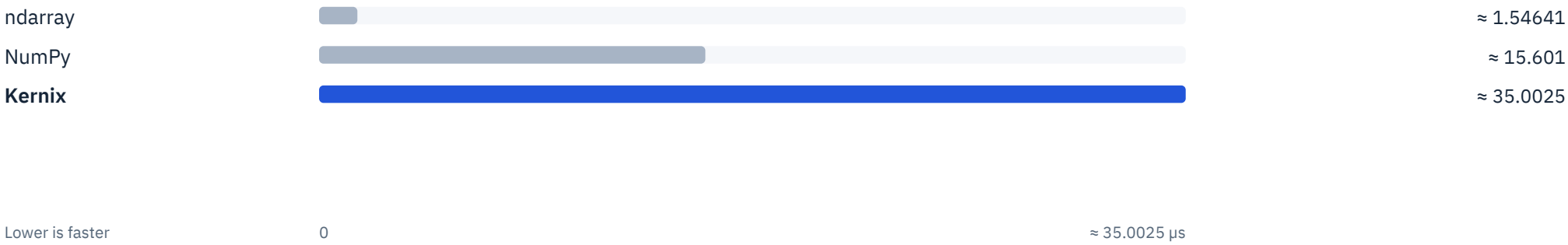
Kernix 0.1.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Kernix 0.1.1 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Elementwise addition

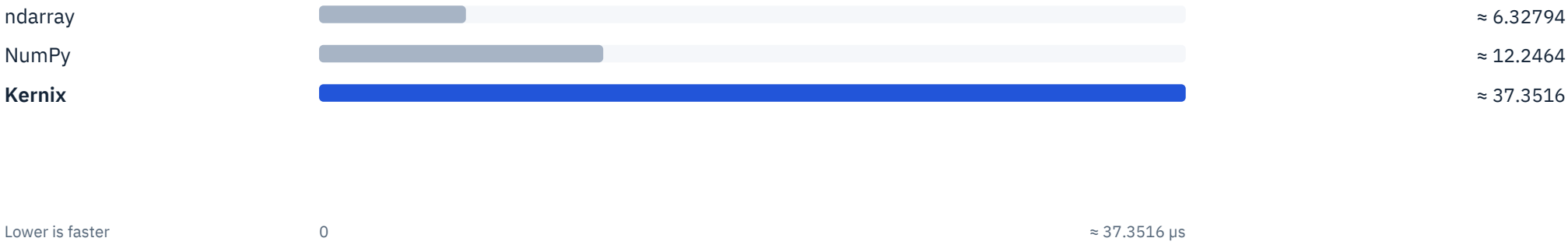
Batch-mean median

65,536 elements · f64 · 1 worker · allocated output

Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



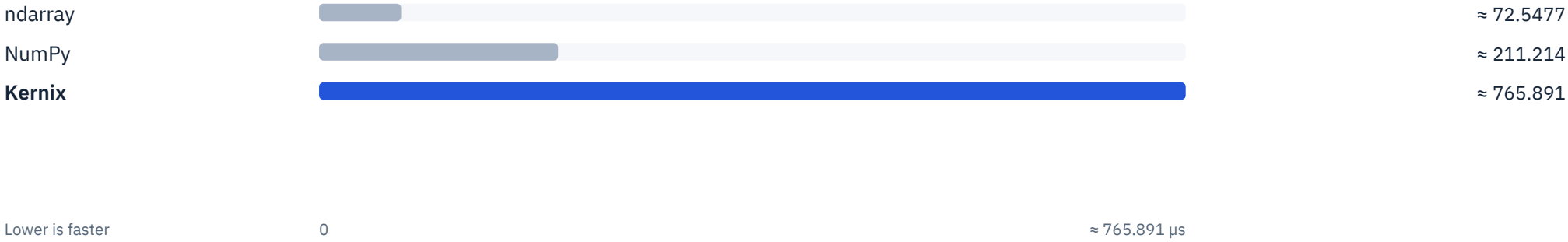
Kernix 0.1.1 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Row softmax

Batch-mean median

256 × 64 · f64 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	35002.51748251748	35.00251748251748
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1546.4088575493288	1.5464088575493288
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	15600.95258889582	15.60095258889582
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	3311847	3311.847
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	88634.74558303886	88.63474558303886
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	1174340.0454545454	1174.3400454545454
Elementwise addition / Kernix 65,536 elements · f64	24624.215551181103	24.624215551181103
Elementwise addition / ndarray 65,536 elements · f64	21962.192273924495	21.962192273924495
Elementwise addition / NumPy 65,536 elements · f64	23282.751396648044	23.282751396648044
Sum reduction / Kernix 65,536 elements · f64	37351.614925373135	37.351614925373135

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sum reduction / ndarray 65,536 elements · f64	6327.939762085548	6.327939762085548
Sum reduction / NumPy 65,536 elements · f64	12246.43095004897	12.24643095004897
Row softmax / Kernix 256 × 64 · f64	765891.3636363636	765.8913636363636
Row softmax / ndarray 256 × 64 · f64	72547.67246376812	72.54767246376812
Row softmax / NumPy 256 × 64 · f64	211214.31092436975	211.21431092436975